

Identifiability in Inverse Reinforcement Learning

Francesco Bellotto, Alex Caldarone, Emilia Farina,
Michele Lanfrancioni, Riccardo Piantoni
Group name: LIRA

EPFL

Introduction

One of the key problems in the field of Reinforcement Learning is the necessity of modelling a reward function for a given environment, from which an optimal behaviour for the agent can be derived effectively. However, due to the ill-conditioning of this problem, designing such a function is often times difficult: either because the problem's characteristics are not known in detail and only a sparse reward function can be crafted, or because it is difficult to parametrise the reward as a combination of a set of chosen features, which is instead particularly useful for interpretability. **Inverse Reinforcement Learning** (IRL) is a novel approach to Reinforcement Learning which aims to leverage observations of other agents in order to recover information on the rewards in these (and other) environments, so that the optimal policy of the agent can then be inferred by employing standard RL algorithms.

Recent literature has been focusing on formally defining the conditions under which the problem is well-posed, i.e. when the reward function can be uniquely identified up to a constant. We define this well-posedness as **Reward Identifiability**. Furthermore, in situations in which the reward function is not identifiable, research has also been trying to circumvent this issue by extending behaviours observed from other agents to the behaviour of an agent in unknown environments.

Preliminaries on Reinforcement Learning

In the context of Reinforcement Learning, a problem is defined as a **Markov Decision Problem**, that is a tuple $(\mathcal{S}, \mathcal{A}, \mathbb{P}, r, \mu, \gamma)$, where:

- $\mathcal{S}$ is the set of all possible states.
- $\mathcal{A}$ is the set of all possible actions.
- For each action a , the matrix P^a with entries $[P^a]_{ss'} = \mathbb{P}(s' | s, a)$ is the transition matrix $(\mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S}))$.
- $r: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function.
- μ is the initial state distribution: $s_0 \sim \mu \in \Delta(\mathcal{S})$.
- γ is the discount factor: $\gamma \in (0, 1)$.

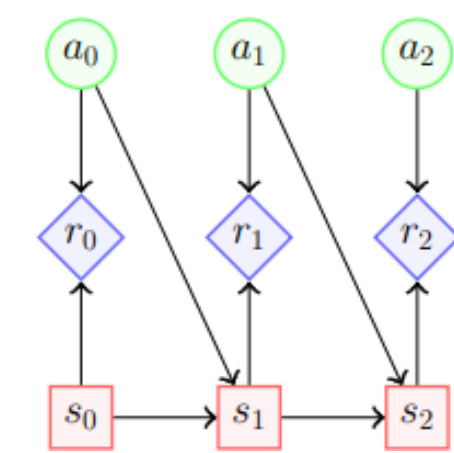


Figure 1. Example of MDP [2]

The RL **Learning Objective** for a classical MDP is defined as

$$J(\pi; r) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{+\infty} \gamma^t r(s_t, a_t) \right].$$

However, it is customary to regularize this objective by adding an entropy term in order to make it smoother and encourage exploration, leading to the **entropy-regularized objective**

$$J_{MaxEnt}(\pi; r) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{+\infty} \gamma^t (r(s_t, a_t) - \lambda \log(\pi(a_t | s_t))) \right].$$

In IRL, we say a MDP is **Weakly Identifiable** if, given the trajectory equivalence $\cong_{\tau}$, the MDP satisfies

$$r \cong_{\tau} \hat{r} \iff p_r \cong_{\tau} p_{\hat{r}}$$

Similarly, a MDP is **Strongly Identifiable** if this relation holds for the state-action equivalence $\cong_{x,a}$.

Finite-horizon and Deterministic Markov Decision Processes

The work [3] provides the necessary and sufficient conditions to determine whether a deterministic MDP model is weakly or strongly identifiable.

Theorem 1: Theorem on Weak Identifiability

A MDP model, under J_{MaxEnt} learning objective, with a domain $d := (\mathcal{S}, \mathcal{A}, P, \mu, \gamma)$ containing deterministic dynamics, is weakly identifiable. On an intuitive level, the optimal trajectory distribution is a noiseless manifestation of the demonstrator's trajectory preferences, which is what allows us to uniquely identify the trajectory level rewards.

In order to determine strong identifiability, we need another condition on our MDP. That is, the MDP has to be proper, i.e. one which yields the same optimal behaviour when the rewards are trajectory equivalent.

A weakly identifiable MDP model is strongly identifiable if and only if for all $r, \hat{r} \in R$, $(r \cong_{\tau} \hat{r}) \implies (r \cong_{x,a} \hat{r})$, i.e. trajectory equivalence implies state-action equivalence.

To characterize strong identifiability, the domain of an MDP is embedded into a graph. We define the **domain graph** for a domain $d := (\mathcal{S}, \mathcal{A}, P, \mu, \gamma)$ to be a tuple $G_d := (V_d, E_d, V_d^0)$, where:

- $V_d := \mathcal{S} \times \mathcal{A}$ are the vertices;
- $V_d^0 := \{(s, a) \mid \mu(s) > 0\}$ are the initial vertices;
- $E_d := \{e := (v, v') \mid ((s, a), (s', a')) \mid v, v' \in V_d, P(s'|s, a) > 0\}$ are the edges.

We define the graph G_d to be strongly connected if there exists a path between any two $v, v' \in V_d$. Further, we say $v \in V_d$ is t -covering, if we can reach all other nodes in G_d in exactly t steps ($t \geq 1$). The graph G_d is t -coverable if an initial vertex $v \in V_d^0$ is t -covering.

Theorem 2: Theorem on Strong Identifiability ([3, Theorem 2])

For all (d, r, T, J) such that the MDP model $\mathcal{P}_{MDP}[R; d, T, J]$ is proper and G_d is strongly connected,

- (Sufficiency) $\mathcal{P}_{MDP}[R; d, T, J]$ is weakly identifiable, G_d is T_0 -coverable, and $T \geq 2T_0 \implies \mathcal{P}_{MDP}[R; d, T, J]$ is strongly identifiable
- (Necessity) $\mathcal{P}_{MDP}[R; d, T, J]$ is strongly identifiable $\implies \mathcal{P}_{MDP}[R; d, T, J]$ is weakly identifiable, G_d is coverable.

That is, *coverability of the domain graph is a necessary and sufficient condition for strong identifiability*.

Characterization of Feasible Reward Functions

Work [1] establishes conditions for the (non-)identifiability of the reward function of entropy-regularized MDPs given one or more optimal policies, for both infinite- and finite-horizon environments. One of the most essential results is the theorem which characterizes the set of all the reward functions that generate an observed policy, supposed optimal:

Theorem 3: Structure of the Feasible Rewards Functions ([1, Theorem 1])

For a fixed policy $\bar{\pi}(a|s) > 0$, discount factor $\gamma \in [0, 1)$, and an arbitrary choice of function $v: \mathcal{S} \rightarrow \mathbb{R}$, there is a unique corresponding reward function

$$r(s, a) = \lambda \log(\bar{\pi}(a|s)) - \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') + v(s)$$

such that the MDP with reward r yields a value function $V_{\lambda}^{\bar{\pi}} = v$ and entropy-regularized optimal policy $\pi_{\lambda}^* = \bar{\pi}$.

Building from this result and the concept of **Value-distinguishing** pair of MDPs, which satisfy

$$w(s) - \gamma \sum_{s' \in \mathcal{S}} P(s'|s, a) v(s') = \tilde{w}(s) - \tilde{\gamma} \sum_{s' \in \mathcal{S}} \tilde{P}(s'|s, a) v(s') \Rightarrow w(s) \text{ or } \tilde{w}(s) \text{ is constant}$$

we conclude the result exposed in **Theorem 2** in the paper, ensuring the uniqueness (or non-existence) of the identified policy up to a constant if the observed policies, for two agents with different transition probabilities and discount factors, are value-distinguishing.

Identifiability and Generalizability from Multiple Experts

The work [4] shows an equivalent identifiability statement, which is easily verifiable and necessary.

Theorem 4: Rank Identifiability Condition

Considering two MDPs on the same sets $\mathcal{S}$ and $\mathcal{A}$, but with different transition dynamics P^1, P^2 and discount factors γ_1, γ_2 . If we observe two experts acting optimally, each in one environment, then the reward function can be recovered up to a constant if and only if

$$\text{rank} \begin{pmatrix} I - \gamma_1 P_{a_1}^1 & I - \gamma_2 P_{a_1}^2 \\ \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & I - \gamma_2 P_{a_{|\mathcal{A}|}}^2 \end{pmatrix} = 2|\mathcal{S}| - 1$$

A possible way to improve reward identifiability is to restrict the class of possible rewards, e.g. by constraining it to be a linear combination of a set of chosen of chosen feature (Feature matching IRL, [4, Theorem 7]).

Theorem 5: Generalizability to Unknown Environments [4, Theorem 11]

$(P^1, \gamma_1), (P^2, \gamma_2)$ generalize to (P^3, γ_3) if and only if

$$\text{rank} \begin{pmatrix} I - \gamma_1 P_{a_1}^1 & I - \gamma_2 P_{a_1}^2 \\ \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & I - \gamma_2 P_{a_{|\mathcal{A}|}}^2 \end{pmatrix} = \text{rank} \begin{pmatrix} I - \gamma_1 P_{a_1}^1 & I - \gamma_2 P_{a_1}^2 & \mathbf{0} \\ \vdots & \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & I - \gamma_2 P_{a_{|\mathcal{A}|}}^2 & \mathbf{0} \\ \vdots & \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & I - \gamma_2 P_{a_{|\mathcal{A}|}}^2 & I - \gamma_3 P_{a_1}^3 \\ \vdots & \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & I - \gamma_2 P_{a_{|\mathcal{A}|}}^2 & I - \gamma_3 P_{a_{|\mathcal{A}|}}^3 \end{pmatrix} - |\mathcal{S}|$$

There exist some cases where identifiability is not possible due the presence of **exogenous variables**, i.e. state variables whose evolution only depends on their previous values, but neither on the other state variables nor the actions taken. However, the generalizability condition in [4, Theorem 11] is often satisfied, even for an environment different from the ones of the observed experts.

Comparisons

- Following from the graph theoretical approach previously introduced, [3] follows up with **Corollary 2** stating that if the MDP is weakly identifiable and the domain graph is aperiodic, then $\exists T_0 \geq 0$ such that $\forall T \geq T_0$, the MDP is strongly identifiable, and if a MDP is strongly identifiable then the MDP is weakly identifiable and the domain graph is aperiodic. [1] adds on to this result by proposing the values of $T_0 = d + 1$ in the case of a domain graph with self loops on the initial state and of $T_0 = d + RR'$ in the case of domain graphs with coprime loops of length R and R' starting at the initial state.
- Furthermore, paper [1] builds on the results in [3] by deriving equivalent conditions which can naturally be extended to settings which are not covered by the latter. For instance, identifiability of the reward function is derived for both the finite- and infinite-horizon cases from the definition in **Theorem 1**; moreover, **Corollary 2** states as an additional condition for identifiability the need for having as many actions as possible future states.
- [4, Theorem 3] provides a condition for reward identification from two experts which is equivalent to [1, Definition 1]. This condition makes it easily verifiable and extendable to various scenarios, e.g. cases where we observe more than two experts, cases where the class of rewards is linearly parametrized by a set of features or cases where we have access to approximated transition matrices.
- [4, Corollary 5] is equivalent to the condition provided by [1, Corollary 3], useful for identifying an action-independent reward from a single expert, in case it exists.

Open Problems

Possible future directions for research in the area are:

- How can we extend the current results to Continuous State Space MDPs?
- Can we recover current results using weaker assumptions on the objective (i.e., not limiting the research to the entropy-regularized objective)?

References

- Haoyang Cao, Samuel N. Cohen, and Lukasz Szpruch. Identifiability in inverse reinforcement learning. *CoRR*, abs/2106.03498, 2021.
- Volkan Cevher. EE-568 Reinforcement Learning, EPFL, Course Slides, 2025.
- Kuno Kim, Shivam Garg, Kirankumar Shiragur, and Stefano Ermon. Reward identification in inverse reinforcement learning. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 5496–5505. PMLR, 18–24 Jul 2021.
- Paul Rolland, Luca Viano, Norman Schuerhoff, Boris Nikolov, and Volkan Cevher. Identifiability and generalizability from multiple experts in inverse reinforcement learning, 2022.